Day 164: The Multiple-Comparisons Problem, with an Honest p-Hacking Demonstration

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 24, Day 4 | Complete study guide

Watch: p-hacking: What it is and how to avoid it! (StatQuest with Josh Starmer) | False Discovery Rates, FDR, clearly explained (StatQuest with Josh Starmer)

Script: `day164_multiple_comparisons.py` · every number in this guide is copied by program from `day164_run_output.txt`, the real output of that script.

One test at the 5% level raises a false alarm 5% of the time when nothing is going on. Run 66 tests and a false alarm becomes the normal outcome. Today cuts Hillstrom's customers into 22 segments and tests 3 outcomes in each, on data where the email made no difference: 0.892 of 1,000 such experiments produced at least one 'significant' finding. Bonferroni, Holm and Benjamini-Hochberg are built from scratch and scored on data where the truth is known, discoveries are checked on fresh customers, and the same trap is shown in machine learning: the best of 100 models looked +0.0034 better than it was.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

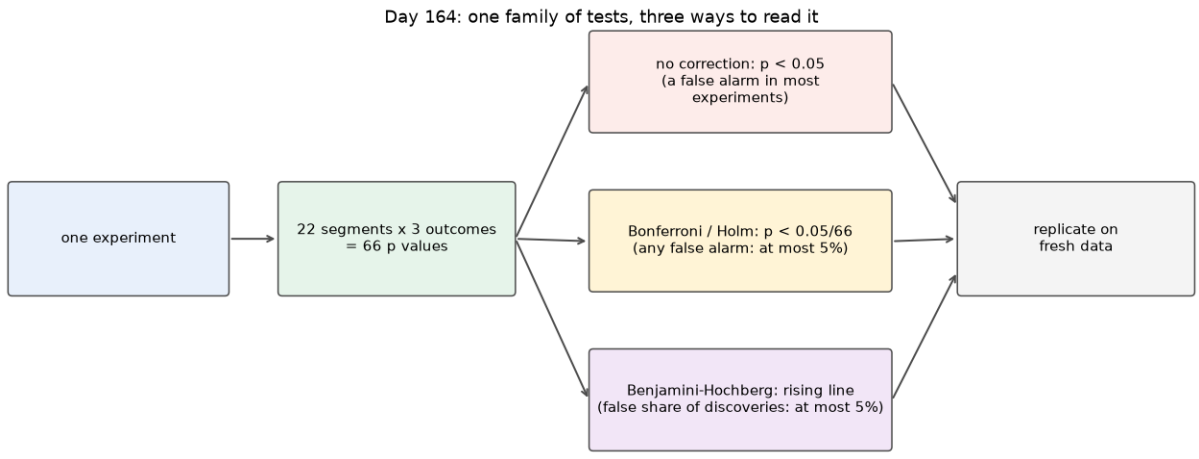
A **family** of tests is every test run while looking for something to report. The **family-wise error rate** (FWER) is the chance that at least one test in the family is a false alarm. The **false discovery rate** (FDR) is the average share of the reported findings that are false.

p-hacking is running many tests (segments, outcomes, time windows, model variants) and reporting only the ones that came out significant, so the 5% promise no longer holds. It does not need bad intent: an analyst who simply keeps looking until something works is doing it.

REAL-WORLD SCENARIO

An email campaign shows no overall effect, so the analyst breaks the results down by region, channel, age of customer and past spend. One slice shows p = 0.001. Is it a real finding about that slice, or the one false alarm that 66 looks were always going to produce?

SYSTEM DIAGRAM



One experiment becomes a family of 66 tests; three ways to decide which count as findings; and the final check that no rule replaces: fresh data.

The data are Day 162's: Hillstrom's 64,000 customers, randomly sent a men's email, a women's email or no email. To make a world where nothing is real, the 21,306 no-email customers are split at random into two fake groups (an **A/A test**): any difference between them is chance by construction. Each comparison is tested in 22 segments (3 channels, 3 zip-code areas, new or returning, 3 kinds of past product, 7 bands of past spend, 4 bands of recency) and on 3 outcomes (visit, purchase, spend): 66 tests. Day 162's two-proportion and Welch tests are imported unchanged.

Today's part for the week's build

`day164_exercise.py` is today's toolkit (the segments, the 66 tests, and Bonferroni, Holm and Benjamini-Hochberg), checked by hand on small lists of p values, including one where Holm finds more than Bonferroni and one where Benjamini-Hochberg's step-up accepts a p value above its own line. It is the fourth part of Day 167's report card.

ACTUAL OUTPUT

```
five p values (0.3, 0.025, 0.005, 0.035, 0.028), m = 5:
  Bonferroni: only p <= 0.05/5 = 0.01, so only 0.005; got [0, 0, 1, 0, 0]
  Holm: 0.005 <= 0.05/5 = 0.01 yes; 0.025 <= 0.05/4 = 0.0125 no, stop; got [0, 0, 1, 0, 0]
  Benjamini-Hochberg: sorted 0.005, 0.025, 0.028, 0.035, 0.3 against 0.01, 0.02, 0.03, 0.04, 0.05;
    0.025 > 0.02 fails its own line, but 0.035 <= 0.04 passes, so the four smallest are all
        rejected; got [0, 1, 1, 1, 1]
three p values (0.01, 0.02, 0.3), m = 3:
  Bonferroni: p <= 0.0167 keeps 0.01 only; got [1, 0, 0]
  Holm: 0.01 <= 0.0167 yes; 0.02 <= 0.05/2 = 0.025 yes; 0.3 <= 0.05 no; got [1, 1, 0]
segments: 22; 22 x 3 outcomes = 66 tests
  channel segments cover 64000 of 64000 customers
  zip_code segments cover 64000 of 64000 customers
  history segments cover 64000 of 64000 customers
  recency segments cover 64000 of 64000 customers
  men's only + women's only + both cover 64000 of 64000 customers
segment_tests on the lesson's first A/A split: 66 p values, smallest 0.0013 (recency 4-6,
    conversion)
```

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. p-hacking on data with no effect

DEFINITION — start here, no prior knowledge assumed

The no-email customers are split once into fake groups A and B, and all 66 tests are run. Nothing differs between A and B except luck, so every 'significant' result is a false alarm. At the 5% level, 66 tests should give about 66 × 0.05 = 3.3 of them on average.

REAL-WORLD SCENARIO

A dashboard with 66 metrics turns red whenever p < 0.05. Even in a week where nothing changed, some tiles go red.

IMPLEMENTATION

```
def segment_tests(d, A, B, S):
    ps, names = [], []
    for sn, sm in S.items():
        a, b = A & sm, B & sm
        for col in OUTCOMES:
            if col == "spend" and d[col][a].max() == 0 and d[col][b].max() == 0:
                ps.append(1.0); EMPTY[0] += 1
            elif col == "spend":
                ps.append(d162.welch_test(d[col][a], d[col][b])[2])
            else:
                ps.append(d162.two_prop_test(d[col][a].sum(), a.sum(), d[col][b].sum(),
                    b.sum())[2])
            names.append(f"{sn}, {col}")
    return np.array(ps), names
```

ACTUAL OUTPUT

```
one A/A split of the 21306 no-email customers into 'A' (10657) and 'B' (10649); 66 tests (22 segments x 3 outcomes)
  'significant' at 0.05: 2 (chance alone expects 3.3):
    recency 4-6, conversion: p 0.0013
    recency 4-6, spend: p 0.0118
  after correction: Bonferroni 0, Holm 0, Benjamini-Hochberg 0
Actual output from this lesson's run.
```

What it means

This split gave 2 'findings', both in the same segment: recency 4-6 purchases (p 0.0013) and spend (p 0.0118). They are not two independent signals: a customer who buys also spends. Written up, this would read as 'customers who last bought 4 to 6 months ago respond differently', a story about nothing. After any of the three corrections, no finding survived.

One split could be lucky or unlucky. The next section repeats it 1,000 times.

WHY CHOOSE THIS?

Looking at many segments is how real effects in small groups are found.

WHY NOT THIS? (TRADEOFFS)

Without a correction, the chance of a false finding grows with every extra look, and the report never says how many looks there were.

2. Bonferroni and Holm: the chance of any false alarm

DEFINITION — start here, no prior knowledge assumed

Bonferroni divides the 5% among the tests: with m tests, each must reach p ≤ 0.05/m. The chance that any of them is a false alarm is then at most 5%, whatever the tests' relationships.

Holm gives the same guarantee but finds more. Sort the p values; the smallest must beat 0.05/m, the next 0.05/(m – 1), and so on, stopping at the first failure. Each step is looser than Bonferroni's single threshold, so Holm never rejects less.

REAL-WORLD SCENARIO

A drug trial with 10 secondary outcomes: regulators require that the chance of claiming any false benefit stays at 5%.

MATHEMATICS

Bonferroni: reject when $p \le \alpha/m$ Holm: sort $p_{(1)} \le \ldots \le p_{(m)}$; reject $p_{(k)}$ while $p_{(j)} \le \alpha/(m - j + 1)$ for every $j \le k$

FWER = P(at least one false rejection); if 66 independent tests: $1 - 0.95^{66}$

IMPLEMENTATION

```
def holm(p, alpha=ALPHA):
    order = np.argsort(p); m = len(p); rej = np.zeros(m, bool)
    for k, i in enumerate(order):
        if p[i] > alpha / (m - k):
            break
        rej[i] = True
    return rej
```

ACTUAL OUTPUT

```
1000 splits, 66 tests each (44 s); family-wise error = share of splits with at least one false discovery
  no correction     : family-wise error 0.892, false discoveries per split 3.13
  Bonferroni        : family-wise error 0.030, false discoveries per split 0.04
  Holm              : family-wise error 0.030, false discoveries per split 0.04
  Benjamini-Hochberg: family-wise error 0.032, false discoveries per split 0.05
if the 66 tests were independent: 1 - 0.95^66 = 0.966; 0 of the 22000 spend tests had no buyer in either group (p = 1)
Actual output from this lesson's run.
```

What it means

Without correction, 0.892 of the 1,000 experiments produced at least one false finding, about 3.13 per experiment. Bonferroni and Holm brought that to 0.030, below the promised 5%; Benjamini-Hochberg gave 0.032. When nothing at all is real, Benjamini-Hochberg also controls the chance of any false alarm, because then every discovery is false.

If the 66 tests were independent, the uncorrected rate would be 0.966. A likely reason it was lower, not tested here: the tests overlap (every customer sits in five segments, and a purchase is also a visit and a spend), so their false alarms tend to arrive together. The same overlap is a likely reason Bonferroni's 0.030 sits below its 0.05 bound: the bound assumes the worst case.

WHY CHOOSE THIS?

A clear guarantee about any false alarm, for any pattern of dependence between the tests; Holm is uniformly better than Bonferroni.

WHY NOT THIS? (TRADEOFFS)

Strict: with many tests, real but modest effects get thrown away (next section).

3. Benjamini-Hochberg, scored where the truth is known

DEFINITION — start here, no prior knowledge assumed

Benjamini-Hochberg (BH) controls a different error: the false share of the discoveries. Sort the p values and draw a rising line k × 0.05/m; find the largest k whose p value is under the line, and reject that one and every smaller one. A p value may sit above its own point on the line and still be rejected, if a larger one further on is under it (the exercise shows this).

To score the rules, the truth must be known. With all 64,000 customers, the men's email against none gives p < 0.0001 in 32 of the 66 segment-outcome tests: those 32 are treated as certainly real. Each of 300 repeats then takes a random 30% of customers and tests those 32 real comparisons together with 66 A/A comparisons (certainly null), and counts what each rule finds.

REAL-WORLD SCENARIO

A screen of 20,000 genes: a biologist will follow up 100 candidates and can accept that 5 of them are duds, but not that none of the real ones make the list.

MATHEMATICS

BH: sort $p_{(1)} \le \ldots \le p_{(m)}$; k^* = the largest k with $p_{(k)} \le k\alpha/m$; reject $p_{(1)}, \ldots, p_{(k^*)}$

FDR = average of (false discoveries / all discoveries), counted as 0 when nothing is discovered; BH keeps it $\le (m_0/m)\alpha$ for independent or positively related tests

IMPLEMENTATION

```
def benjamini_hochberg(p, alpha=ALPHA):
    order = np.argsort(p); m = len(p)
    ok = p[order] <= alpha * np.arange(1, m + 1) / m
    k = np.flatnonzero(ok).max() + 1 if ok.any() else 0
    rej = np.zeros(m, bool); rej[order[:k]] = True
    return rej
```

ACTUAL OUTPUT

```
truth from all 64,000 customers: men's email vs none is certainly real (p < 0.0001) in 32 of 66 segment x outcome tests; the
    other 34 are left out (truth uncertain)
each repeat: a random 30% of customers; 32 real tests (men's email vs none) plus 66 null tests (an A/A split of the no-email
    customers)
300 repeats (18 s); FDR = mean share of discoveries that are false; FWER = share of repeats with any
  no correction     : real effects found 29.35 of 32, false discoveries 2.70, FDR 0.0790, FWER 0.790
  Bonferroni        : real effects found 21.15 of 32, false discoveries 0.01, FDR 0.0003, FWER 0.003
  Holm              : real effects found 21.44 of 32, false discoveries 0.01, FDR 0.0004, FWER 0.007
  Benjamini-Hochberg: real effects found 26.70 of 32, false discoveries 0.61, FDR 0.0210, FWER 0.350
(171 of the 13200 spend tests had no buyer in either group and were given p = 1)
Actual output from this lesson's run.
```

What it means

Without correction, 29.35 of the 32 real effects were found per repeat, but with 2.70 false discoveries and a chance of any false alarm of 0.790. Bonferroni and Holm cut false discoveries to 0.01 per repeat but found only 21.15 and 21.44 real effects. BH found 26.70, about five more, at the cost of 0.61 false discoveries per repeat: its false discovery rate was 0.0210, within its promise (at most 66/98 × 0.05 = 0.034 here, since 66 of the 98 tests are null).

BH's chance of any false alarm was 0.350: it does not promise to control that, and it did not. Choose the error that matches the decision: any false alarm is costly (a regulatory claim) → Holm; a shortlist that will be checked again → BH. A caution on the benchmark: 'real' was decided on all customers, which include each repeat's sample; with p < 0.0001 on 64,000 customers these effects are very unlikely to be false, but the truth is estimated, not designed. (171 spend tests had no buyer in either group; they were given p = 1.)

WHY CHOOSE THIS?

Finds many more real effects than Holm while keeping the false share of discoveries near its target.

WHY NOT THIS? (TRADEOFFS)

Makes no promise about any single false alarm: in a third of repeats here at least one discovery was false.

4. Does a discovery replicate on fresh customers?

DEFINITION — start here, no prior knowledge assumed

No correction is as convincing as seeing a finding again on data that played no part in finding it. The customers are split in half at random. In the first half, all 66 tests are run and the best-looking one is picked; then the same test is run on the second half. This is done 200 times in a world with no effect (A/A) and in one with a real effect (men's email against none).

REAL-WORLD SCENARIO

A retailer finds that a promotion worked best in one region. Before rolling it out there, it reruns the promotion in that region next month.

ACTUAL OUTPUT

```
  A/A (no effect): over 200 repeats
    the best of 66 in the discovery half: median p 1.9e-02, below 0.05 in 0.855 of repeats; the same test on fresh
        customers: below 0.05 in 0.045
    every test below 0.05 in the discovery half (2.8 per repeat): 0.052 of them below 0.05 again on fresh customers
  men's email vs none: over 200 repeats
    the best of 66 in the discovery half: median p 2.8e-34, below 0.05 in 1.000 of repeats; the same test on fresh
        customers: below 0.05 in 1.000
    every test below 0.05 in the discovery half (48.8 per repeat): 0.776 of them below 0.05 again on fresh customers
(60 of the 17600 spend tests here had no buyer in either group and were given p = 1)
Actual output from this lesson's run.
```

What it means

With no effect, the best of 66 was 'significant' in 0.855 of repeats, and on fresh customers the same test was significant in only 0.045: the rate of a single honest test. Across all discoveries, 0.052 replicated. With the real email effect, the winner replicated every time (1.000), and 0.776 of all discoveries did. A likely reason that is not higher, not tested here: some real effects are small, and half the customers is too few to detect them every time.

Replication separates the two worlds far more sharply than any p value in the discovery data could.

WHY CHOOSE THIS?

The strongest check there is, and it needs no assumptions about how the tests relate.

WHY NOT THIS? (TRADEOFFS)

Costs half the data (or a second experiment), and a real but small effect can fail to replicate by bad luck.

5. The same trap in ML: the best of 100 models

DEFINITION — start here, no prior knowledge assumed

Choosing among many models on one validation set is a multiple-comparisons problem too. Here 100 logistic regressions are trained on Day 161's UCI Adult fitting rows, each on a random half of the 12 features. The validation rows are split in half at random: the model with the best accuracy on the first half is chosen, then scored on the second half. This is repeated over 500 random halvings.

REAL-WORLD SCENARIO

A Kaggle team submits its best of 100 variants and is surprised when the private leaderboard score comes out lower than the public one.

IMPLEMENTATION

```
gaps = []; pick = []
for s in range(S_SPLIT):
    h = rng.rand(len(va)) < 0.5
    a, b = C[:, h].mean(1), C[:, ~h].mean(1)
    w = np.argmax(a); gaps.append(a[w] - b[w]); pick.append(w)
gaps = np.array(gaps)
```

ACTUAL OUTPUT

```
100 logistic regressions, each on a random half of the 12 features (15 s); validation accuracy 0.7512 to 0.8394
choose the best on one half of the validation rows, score it on the other half (500 random halvings):
  the winner's accuracy drops by +0.0034 on average (spread of that mean 0.0004); 6 different models won at least once
  for comparison, a model fixed in advance drops by -0.0001
Actual output from this lesson's run.
```

What it means

Validation accuracy ranged from 0.7512 to 0.8394. The chosen model's accuracy dropped by +0.0034 on average when scored on rows that played no part in choosing it (the uncertainty of that average is 0.0004). A model fixed in advance dropped by -0.0001: no drop, as it should be. Only 6 different models ever won: the top few were close, and luck on the selection rows decided among them.

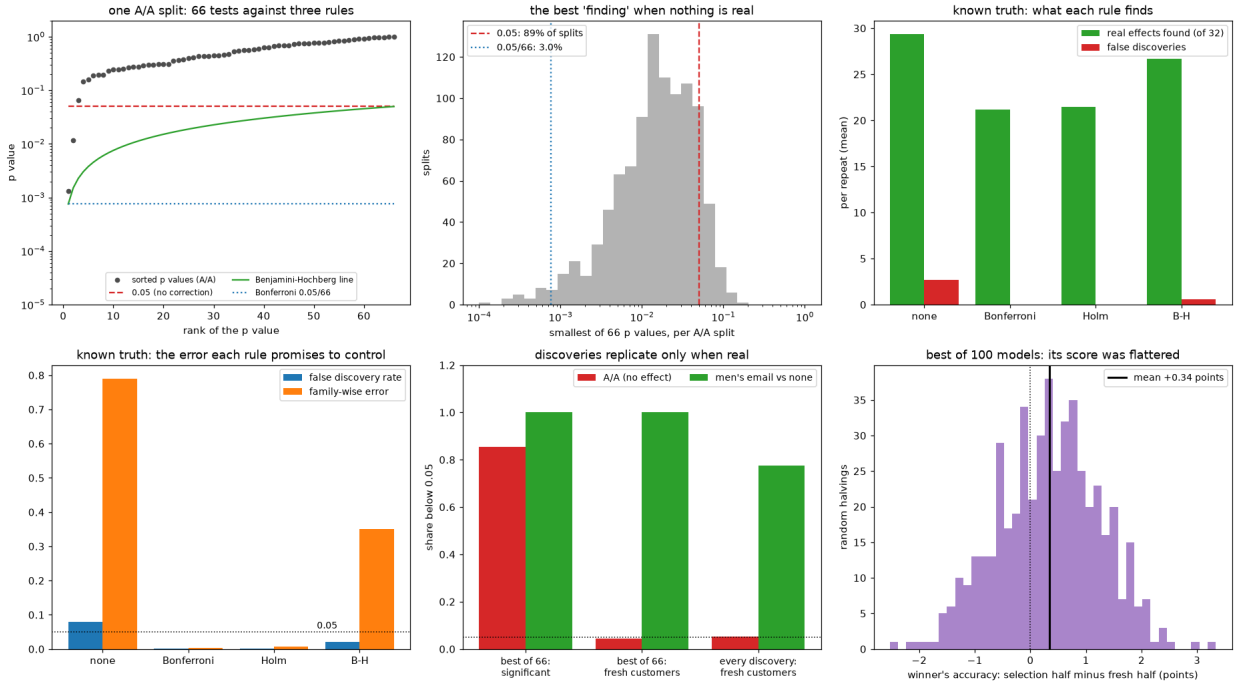
A third of a point is small, but it is systematic, and it grows with the number of models tried. It is why a final score must come from a test set used once, after every choice is made: Day 167 does exactly that with Adult's test file.

WHY CHOOSE THIS?

Picking the best model on validation data is still the right way to choose.

WHY NOT THIS? (TRADEOFFS)

The winner's validation score is flattered; report it from untouched test data instead.



Top: one A/A split's 66 p values against the three rules; the smallest p value of each of 1,000 A/A experiments; what each rule finds when the truth is known. Bottom: the error each rule controls; replication on fresh customers; how much the best of 100 models was flattered.

Interview preparation — technical questions

Q: You test 66 segments and one gives p = 0.001. What do you do?

A: Correct for the 66 tests (Holm: 0.001 is above 0.05/66 = 0.00076, so not significant), then confirm it on fresh data.

Q: Bonferroni or Holm?

A: Holm: the same guarantee, never fewer discoveries. Here 21.44 real effects found against 21.15.

Q: When is Benjamini-Hochberg the right choice?

A: When the findings form a shortlist that will be checked again: it found 26.70 of 32 real effects against Holm's 21.44, with 2.1% of discoveries false.

Q: How would you check a correction method?

A: On A/A splits, where every discovery is false by construction, and on a mix of known real and null effects.

Q: Why does the best validation score overstate a model?

A: It was chosen for being high, partly by luck: here the winner of 100 dropped by 0.0034 on fresh rows.

Interview preparation — theoretical questions

Q: What is the difference between FWER and FDR?

A: FWER: the chance of even one false discovery. FDR: the average share of discoveries that are false. BH controls FDR and had an FWER of 0.350 here.

Q: Why was the uncorrected false-alarm rate 0.892, not 0.966?

A: The 66 tests overlap and are correlated; a likely reason, not tested here, is that their false alarms arrive together.

Q: Is p-hacking always deliberate?

A: No: any analysis that tries several cuts and reports the best one does it, whatever the intent.

Q: What does pre-registration fix?

A: It fixes the family in advance, so the number of tests is known and a correction can be applied honestly.

Key takeaway and what's next

> **KEY TAKEAWAY**
>
> A p value is honest only for a test chosen in advance. Look at 66 and the best of them is significant most of the time even when nothing is real. Holm keeps the chance of any false alarm at 5%; Benjamini-Hochberg keeps the false share of discoveries at 5% and finds more; fresh data is the final check, and it applies to choosing models too.

- No effect, 66 tests: at least one false finding in 0.892 of experiments; 0.030 after Holm.
- Known truth: BH found 26.70 of 32 real effects, Holm 21.44; BH's FDR 0.0210.
- Replication with no effect: 0.045; with a real effect: 1.000.
- Best of 100 models: flattered by +0.0034 accuracy.

Suggested watch order, if starting from zero

- 1. p-hacking: What it is and how to avoid it! (StatQuest with Josh Starmer) — how p-hacking happens, in pictures.
- 2. False Discovery Rates, FDR, clearly explained (StatQuest with Josh Starmer) — the false discovery rate and the Benjamini-Hochberg method.

Day 165 preview

- **Calibration.** When a model says 70%, does the event happen 70% of the time? Reliability diagrams for Day 161's models, the expected calibration error, and two fixes (Platt scaling and isotonic regression) fitted on held-out rows.